Part II: Modular Arithmetic Solutions

Problem 2

Prove that

$$ab \bmod n = \big((a \bmod n)(b \bmod n)\big) \bmod n.$$

By the division algorithm, there exist integers q_1, q_2 and remainders r, s such that

$$a = q_1 n + r, \qquad 0 \le r < n,$$

and

$$b = q_2 n + s, \qquad 0 \le s < n.$$

By definition,

$$r = a \bmod n \qquad \text{and} \qquad s = b \bmod n.$$

Now multiply the two expressions:

$$\begin{aligned} ab &= (q_1 n + r)(q_2 n + s) \\ &= q_1 q_2 n^2 + q_1 n s + q_2 n r + rs \\ &= n(q_1 q_2 n + q_1 s + q_2 r) + rs. \end{aligned}$$

The first term is a multiple of n. Therefore,

$$ab \equiv rs \pmod{n}.$$

Congruent integers have the same remainder modulo n, so

$$ab \bmod n = rs \bmod n.$$

Since $r = a \bmod n$ and $s = b \bmod n$, we obtain

$$\boxed{ab \bmod n = \big((a \bmod n)(b \bmod n)\big) \bmod n}.$$

Problem 3

We use the convention that $x \bmod n$ is the unique integer r satisfying

$$x = qn + r, \qquad 0 \le r < n.$$

(a) 9716 mod 87

Divide 9716 by 87:

$$87(111) = 9657.$$

Thus

$$9716 = 87(111) + 59,$$

where $0 \leq 59 < 87$. Therefore,

$$\boxed{9716 \bmod 87 = 59}.$$

(b) -9716 mod 87

From the previous calculation,

$$9716 = 87(111) + 59.$$

Negating both sides gives

$$-9716 = -87(111) - 59.$$

The number -59 is not an allowed standard remainder because it is negative. Rewrite it as

$$-59 = -87 + 28.$$

Therefore,

$$\begin{aligned} -9716 &= -87(111) - 87 + 28 \\ &= 87(-112) + 28. \end{aligned}$$

Since $0 \leq 28 < 87$,

$$\boxed{-9716 \bmod 87 = 28}.$$

As a check,

$$28 = 87 - 59.$$

(c) 87 mod 9716

Because

$$0 \leq 87 < 9716,$$

division by 9716 gives

$$87 = 9716(0) + 87.$$

Hence,

$$\boxed{87 \bmod 9716 = 87}.$$

(d) $-87 \bmod 9716$

We need to express -87 with a nonnegative remainder less than 9716:

$$-87 = 9716(-1) + (9716 - 87).$$

Since

$$9716 - 87 = 9629,$$

we have

$$-87 = 9716(-1) + 9629.$$

Because $0 \leq 9629 < 9716$,

$$\boxed{-87 \bmod 9716 = 9629}.$$

Therefore, the four answers are

$$\boxed{59, \quad 28, \quad 87, \quad 9629}.$$

Problem 4

Consider

$$\mathbb{Z}_{27} = \{0, 1, 2, \ldots, 26\}.$$

An element $a \in \mathbb{Z}_{27}$ has a multiplicative inverse modulo 27 if there exists $x \in \mathbb{Z}_{27}$ such that

$$ax \equiv 1 \pmod{27}.$$

A standard result states that a has a multiplicative inverse modulo 27 if and only if

$$\gcd(a, 27) = 1.$$

Since

$$27 = 3^3,$$

an integer is relatively prime to 27 exactly when it is not divisible by 3. Thus the invertible elements are

$$1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26.$$

We now find the inverse of each element.

Finding the inverse pairs

$$1 \cdot 1 = 1 \equiv 1 \pmod{27},$$

so $1^{-1} \equiv 1 \pmod{27}$.

$$2 \cdot 14 = 28 = 27 + 1 \equiv 1 \pmod{27},$$

so $2^{-1} \equiv 14 \pmod{27}$, and therefore $14^{-1} \equiv 2 \pmod{27}$.

$$4 \cdot 7 = 28 = 27 + 1 \equiv 1 \pmod{27},$$

so $4^{-1} \equiv 7 \pmod{27}$, and therefore $7^{-1} \equiv 4 \pmod{27}$.

$$5 \cdot 11 = 55 = 2(27) + 1 \equiv 1 \pmod{27},$$

so $5^{-1} \equiv 11 \pmod{27}$, and therefore $11^{-1} \equiv 5 \pmod{27}$.

$$8 \cdot 17 = 136 = 5(27) + 1 \equiv 1 \pmod{27},$$

so $8^{-1} \equiv 17 \pmod{27}$, and therefore $17^{-1} \equiv 8 \pmod{27}$.

$$10 \cdot 19 = 190 = 7(27) + 1 \equiv 1 \pmod{27},$$

so $10^{-1} \equiv 19 \pmod{27}$, and therefore $19^{-1} \equiv 10 \pmod{27}$.

$$13 \cdot 25 = 325 = 12(27) + 1 \equiv 1 \pmod{27},$$

so $13^{-1} \equiv 25 \pmod{27}$, and therefore $25^{-1} \equiv 13 \pmod{27}$.

$$16 \cdot 22 = 352 = 13(27) + 1 \equiv 1 \pmod{27},$$

so $16^{-1} \equiv 22 \pmod{27}$, and therefore $22^{-1} \equiv 16 \pmod{27}$.

$$20 \cdot 23 = 460 = 17(27) + 1 \equiv 1 \pmod{27},$$

so $20^{-1} \equiv 23 \pmod{27}$, and therefore $23^{-1} \equiv 20 \pmod{27}$.

$$26 \cdot 26 = 676 = 25(27) + 1 \equiv 1 \pmod{27},$$

so $26^{-1} \equiv 26 \pmod{27}$.

Complete table

a	1	2	4	5	7	8	10	11	13	14	16	17	19	20	22	23	25	26
$a^{-1} \bmod 27$	1	14	7	11	4	17	19	5	25	2	22	8	10	23	16	20	13	26

The inverse pairs are

$$(1,1),\ (2,14),\ (4,7),\ (5,11),\ (8,17),\ (10,19),$$

$$(13,25),\ (16,22),\ (20,23),\ (26,26).$$

Observation

All the elements with multiplicative inverses have the following property:

$$\boxed{\gcd(a,27)=1.}$$

Equivalently, none of them is divisible by 3. The elements that do *not* have inverses are

$$0,3,6,9,12,15,18,21,24,$$

because each shares a factor of 3 with 27.

There are 18 invertible elements. This also agrees with Euler's totient function:

$$\varphi(27)=27\left(1-\frac{1}{3}\right)=18.$$